Pipe Bend Thickness Report

ASME B31.3-2018 para. 304.2.1 | Pipe bend under internal pressure

Project	QA — مثال كوع
Prepared by	Device preview
Design reference	B31.3-2018 QA
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01 BEND GEOMETRY

Material / grade	User coefficients - example only
Outside diameter D	114.3 mm
Bend radius to centreline R1	171.45 mm (1.5 D)
Inspection / pipe reference	QA-UT

02 DESIGN INPUTS

Internal design gauge pressure P	20 bar(g) = 2 MPa
Allowable stress S	100 MPa
Quality factor E / weld factor W	0.85 / 0.9
Coefficient Y	0.4
Corrosion / erosion allowance	1.5 / 0.5 mm
Mechanical removal allowance	1 mm (before fabrication)

03 CALCULATION TRACE

t = P x D / [2 x ((S x E x W / I) + P x Y)], where I depends on the bend geometry: I_intrados = [4(R1/D) - 1] / [4(R1/D) - 2] and I_extrados = [4(R1/D) + 1] / [4(R1/D) + 2]. I_intrados is always above 1 and governs.

Intrados factor I	1.25
Extrados factor I	0.875
Pressure thickness at intrados	1.8435 mm
Pressure thickness at extrados	1.2955 mm
Matching straight pipe, I = 1	1.4787 mm
Bend wall before fabrication	4.8435 mm
Required remaining wall at intrados	3.8435 mm

04 MEASURED INTRADOS WALL

Minimum measured intrados wall	4 mm
Required remaining wall	3.8435 mm

Measured minus required	**0.1565 mm**

MEETS the internal-pressure wall requirement for these inputs.

05 LIMITS AND REFERENCES

Basis: ASME B31.3-2018, 304.2.1. S, E, W, Y and the allowances are user inputs and require verification against the code and the project specification. Bending thins the wall at the intrados; this report checks the required thickness at each point of the bend rather than computing the geometric thinning, for which the code gives no equation. Miter bends (304.2.3), external pressure, other sustained or occasional loads, fatigue, and remaining-life assessment are outside this report.